

Ques I: let $x[n]$ be the original, $y[n]$ the transmitted and $z[n]$ the reconstructed signal.

Then $y[n] = x\left[\frac{3n}{2}\right]$ [downsample the signal by factor of $\frac{3}{2}$:
use linear-interpolation for the zero sample points]

and $z[n] = y\left[\frac{2}{3}n\right]$ [up-sample the signal by factor of $\frac{3}{2}$]

Ques II:

1. $x[n] = \{0, 0.5, 0.87, 1, 0.87, 0.5\}$

$$x\left[\frac{n}{2}\right] = \{0, 0, 0.5, 0, 0.87, 0, 1, 0, 0.87, 0, 0.5\}$$

↓ interpolate

$$\{0, 0.25, 0.5, 0.685, 0.87, 0.935, 1, 0.935, 0.87, 0.685, 0.5\}$$

$$\therefore x\left[\frac{3n}{2}\right] = \{0.0, 0.685, 1, 0.685\}$$

$$y[n] = x\left[\frac{3n}{2}\right] = \text{input signal to the channel.}$$

2. $y[n] = \{0, 0.685, 1, 0.685\}$

$$y\left[\frac{n}{3}\right] = \{0, 0, 0, 0.685, 0, 0, 1, 0, 0, 0.685\}$$

↓ interpolate

$$\{0, 0.228, 0.457, 0.685, 0.790, 0.895, 1, 0.895, 0.790, 0.685\}$$

$$y\left[\frac{2n}{3}\right] = \{0, 0.457, 0.790, 1, 0.790, \}$$

$$\approx \{0, 0.457, 0.790, 1, 0.790, 0.457\}$$

$$z[n] = y\left[\frac{2n}{3}\right] = \text{reconstructed signal.} \quad \xrightarrow{\text{determined through symmetry.}}$$

Q1

b \ a	1	0	1	1
1	1	0	1	1
1	1	0	1	1
0	0	0	0	0
1	1	0	1	1

$$a[n] * b[n] = \{1, 1, 1, 3, 1, 1, 1\} = x[n]$$

c \ b	1	1	0	1
1	1	1	0	1
1	1	1	0	1
1	1	1	0	1
1	1	1	0	1

$$b[n] * c[n] = \{1, 2, 2, 3, 2, 1, 1\}$$

c \ a	1	0	1	1
1	1	0	1	1
1	1	0	1	1
1	1	0	1	1
1	1	0	1	1

$$a[n] * c[n] = \{1, 1, 2, 3, 2, 2, 1\} = y[n]$$

a \ a	1	0	1	1
1	1	0	1	1
0	0	0	0	0
1	1	0	1	1
1	1	0	1	1

$$a[n] * a[n] = \{1, 0, 2, 2, 1, 2, 1\}$$

b \ b	1	1	0	1
1	1	1	0	1
1	1	1	0	1
0	0	0	0	0
1	1	1	0	1

$$b[n] * b[n] = \{1, 2, 1, 2, 2, 0, 1\} = z[n]$$

$$\boxed{5} \quad \text{Ev} \{P(x)\} = \frac{P(x) + P(-x)}{2}$$

$$3 \cos(x) - 5x^2$$

$$\text{Odd} \{P(x)\} = \frac{P(x) - P(-x)}{2} = x^3$$

$$+ \sin(2x)$$

$$\text{Ev} \{P(x)\} + \text{Odd} \{P(x)\} = P(x) \quad [\text{verified}]$$

Ques-2

1. $B_n = 0$ [even signal]
2. $A_n = 0$ [odd signal]

$$\begin{aligned} \boxed{6} \quad X(j\omega) &= \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt \\ &= \int_{-\infty}^{+\infty} e^{-t} e^{-j\omega t} dt \\ &= e^{-\omega/4} \int_{-\infty}^{+\infty} e^{-(t + \frac{j\omega}{2})} dt \\ &= \sqrt{\pi} e^{-\omega/4} \quad \left[\because \int_{-\infty}^{+\infty} e^{-x^2} dx = \sqrt{\pi} \right] \end{aligned}$$

Now,
Energy in time domain, $E_t = \int_{-\infty}^{+\infty} |x(t)|^2 dt$

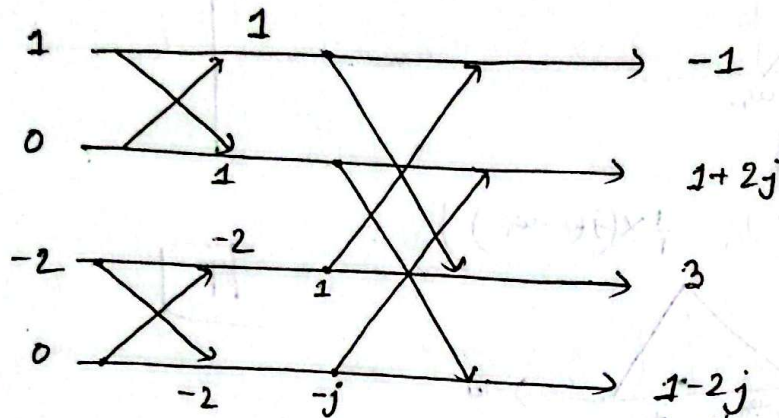
$$\begin{aligned} &= \int_{-\infty}^{+\infty} (e^{-t})^2 dt \\ &= \int_{-\infty}^{+\infty} e^{-(\sqrt{2}t)} dt = \sqrt{\frac{\pi}{2}} \end{aligned}$$

Energy in freq. domain, $E_\omega = \frac{1}{2\pi} \int_{-\infty}^{+\infty} |X(j\omega)|^2 d\omega$

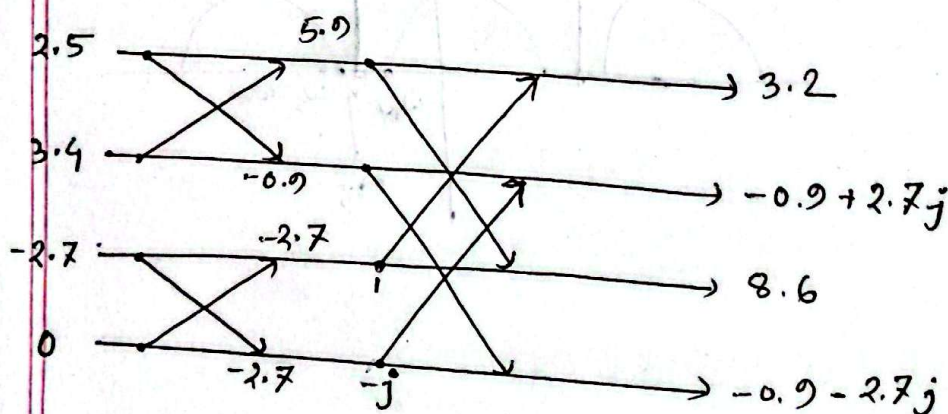
$$\begin{aligned} &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} (\sqrt{\pi} e^{-\omega/4})^2 d\omega \\ &= \frac{1}{2} \int_{-\infty}^{+\infty} e^{-(\omega/\sqrt{2})} d\omega \\ &= \frac{1}{2} \sqrt{2\pi} = \frac{\sqrt{\pi}}{\sqrt{2}} \quad \therefore E_t = E_\omega \end{aligned}$$

7. (a) sample lag = 1 [out of phase; $\therefore$ maximum negative val.]

(b) $x[n] = \{1, -2, 0, 0\} \xrightarrow{\text{FFT}} X[k] = \{-1, 1+2j, 3, 1-2j\}$



$y[n] = \{2.5, -2.7, 3.4, 0\} \xrightarrow{\text{FFT}} Y[k] = \{3.2, -0.9+2.7j, 8.6, -0.9-2.7j\}$



$\therefore X[k] * Y[k] = \{-3.2, -6.3+0.9j, 25.8, -6.3-0.9j\}$

8. (b) $x[n] = \begin{cases} 1, & \text{if } n \leq -1 \\ 0, & \text{otherwise} \end{cases}$

